

DATA Deliverable 3

2026-08-04

```
library(dplyr)
```

```
##
```

```
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':
```

```
##
```

```
## filter, lag
```

```
## The following objects are masked from 'package:base':
```

```
##
```

```
## intersect, setdiff, setequal, union
```

```
library(readr)
```

```
library(ggplot2)
```

```
dataset = read.csv("listings.csv")
```

```
chch_data = dataset %>%
```

```
  filter(neighbourhood_group == "Christchurch City")
```

```
ggplot(chch_data, aes(x = price)) +
```

```
  geom_histogram(binwidth = 10, colour = "white") +
```

```
  labs(
```

```
    title = "House Prices in Christchurch City",
```

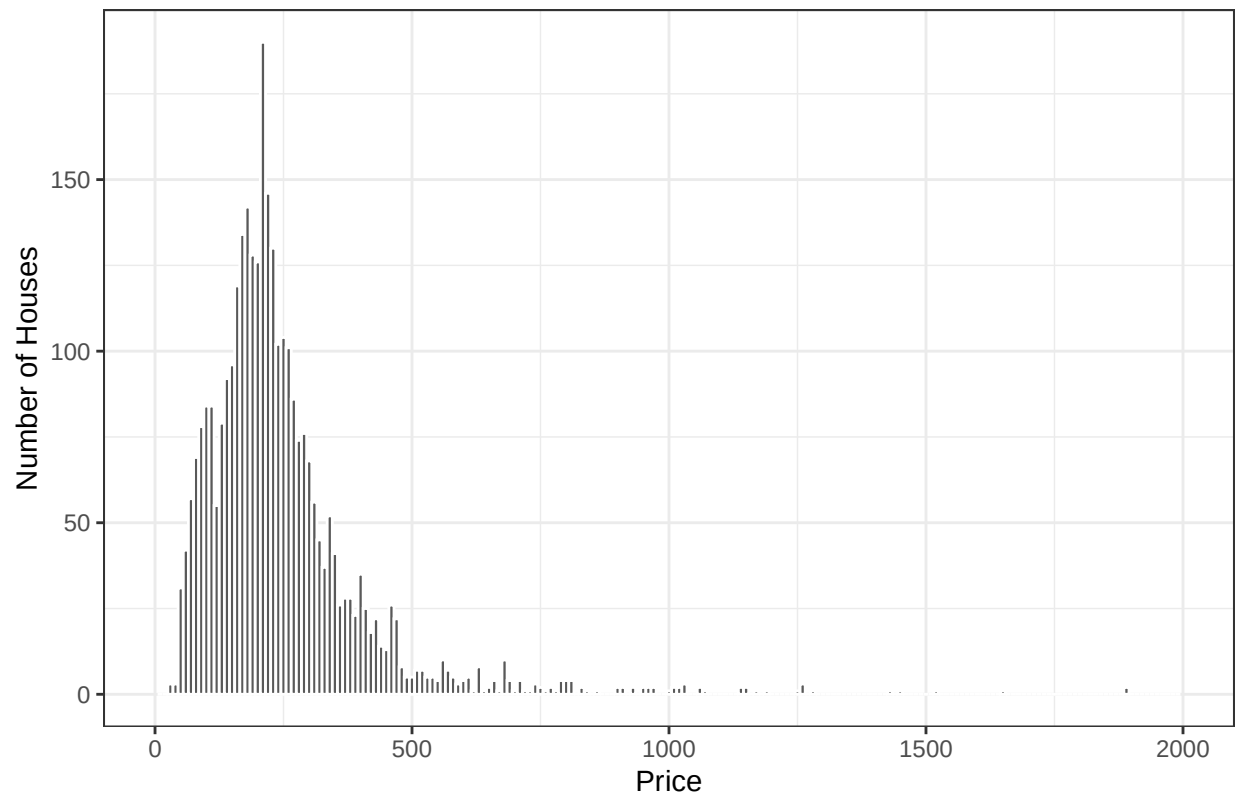
```
    x = "Price",
```

```
    y = "Number of Houses"
```

```
  ) +
```

```
  lims(x = c(0,2000)) + theme_bw()
```

House Prices in Christchurch City

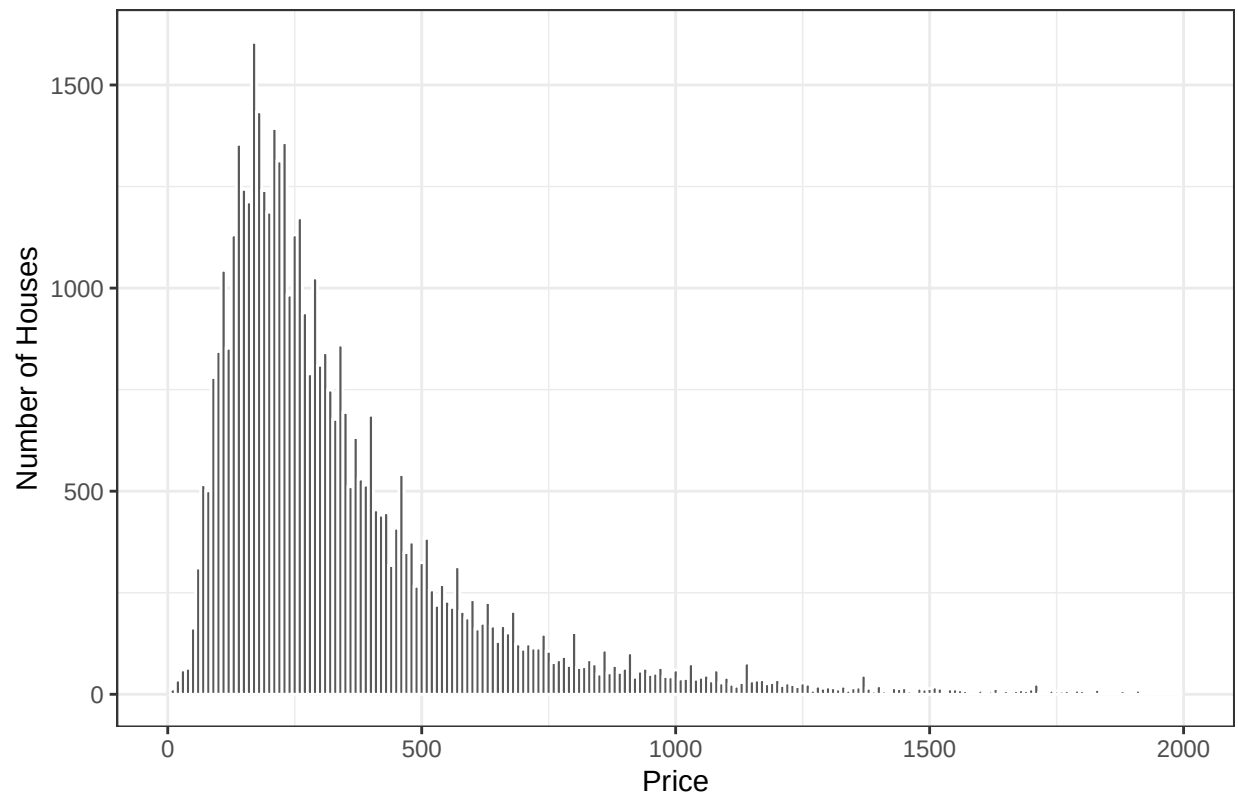


```
dataset = read.csv("listings.csv")

chch_data = dataset %>%
  filter(neighbourhood_group == "Christchurch City")

ggplot(dataset, aes(x = price)) +
  geom_histogram(binwidth = 10, colour = "white") +
  labs(
    title = "House Prices in New Zealand",
    x = "Price",
    y = "Number of Houses"
  ) +
  lims(x = c(0,2000)) + theme_bw()
```

House Prices in New Zealand



```
dataset = read.csv("listings.csv")

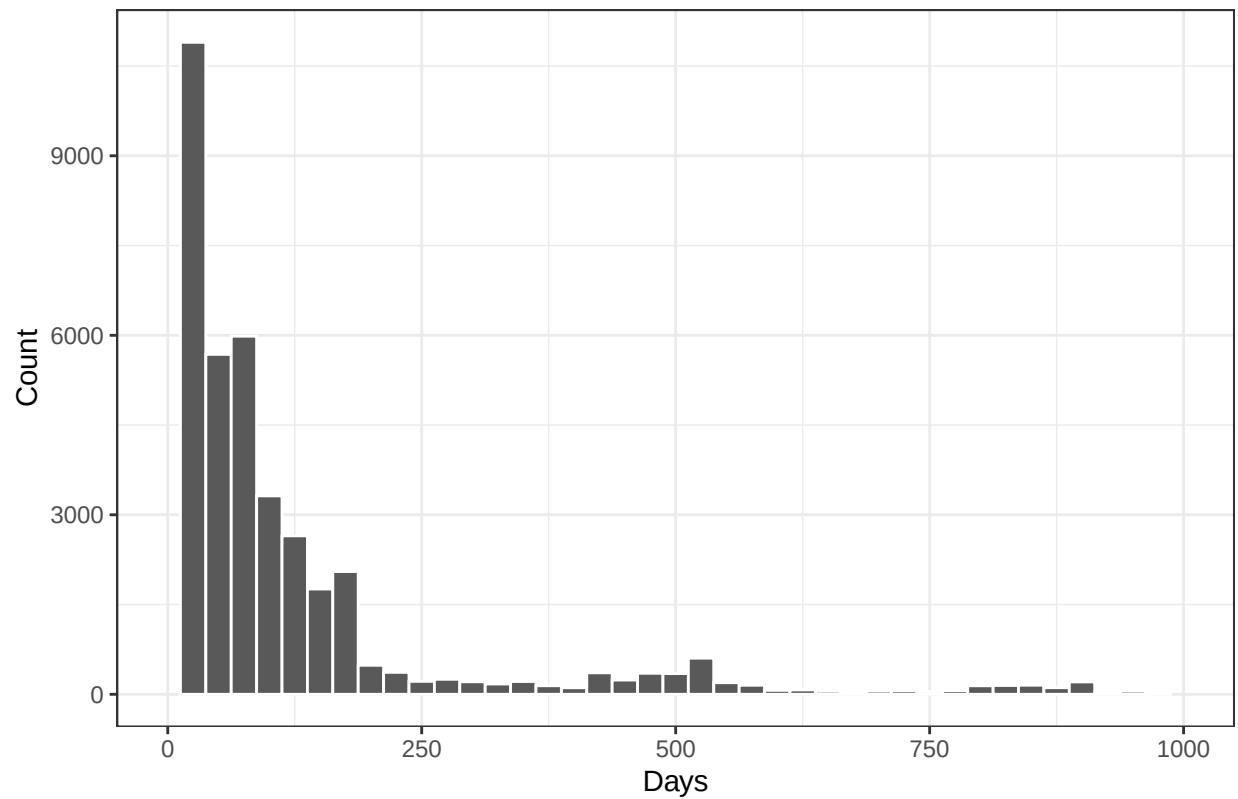
dataset$last_review <- as.Date(dataset$last_review, format = "%Y-%m-%d")

dataset$days_since_june19 <- as.Date("2026-06-19") - dataset$last_review

dataset$days_since_june19 <- as.numeric(dataset$days_since_june19)

ggplot(dataset, aes(x = days_since_june19)) +
  geom_histogram(colour = "white", binwidth = 25) +
  labs(
    title = "Difference Between Last Review and Publish Date in Days",
    x = "Days", y = "Count"
  ) +
  lims(x = c(0, 1000)) + theme_bw()
```

Difference Between Last Review and Publish Date in Days



Number of Top 10% Properties in Christchurch New Zealand

```
dataset = read.csv("listings.csv")

threshold <- quantile(dataset$number_of_reviews, 0.9, na.rm = TRUE)

top_10 <- dataset %>%
  filter(number_of_reviews >= threshold)

top_10 %>%
  filter(neighbourhood_group == "Christchurch City") %>%
  nrow()
```

```
## [1] 343
```